Heterogeneous Graph Attention Network for semi-supervised News Classification

Thesis Mid-Term Presentation

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Introduction

- News classification is one of the application of text classification
- It helps to categorize news article into certain class, so it is useful when accessing relevant information or recommending news to user
- Classification problem deals with both understanding the semantics and grouping the text to its class
- Large amount of label data are usually required for classification problem

Problem Statement

- Difficult to find labeled data and less labeled data usually doesn’t give high accuracy

- Since there is large amount of unlabelled data, getting information from such data will be very beneficial

- Extracting information from less amount of data is also challenging

Objectives

1. To leverage the use of less amount of labelled data as well as large unlabelled data.

2. To convert raw text data to rich semantic heterogeneous graph network.

3. To classify the short news data using semi-supervised HGAT method enhancing the embedding approach

Literature Review

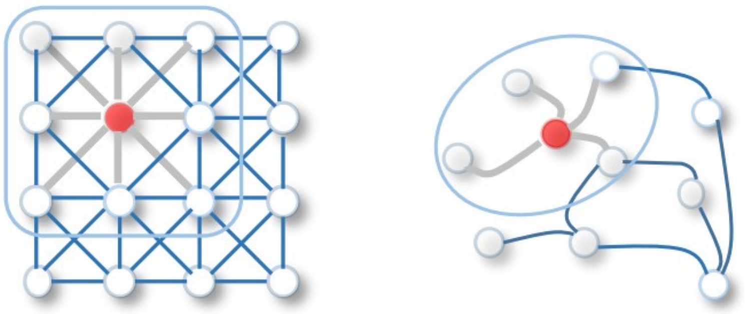
Traditional text classification methods such as SVM (Blei et al., 2003)need a feature engineering step for text representation.

Similarly, various deep neural network such as RNNs (Liu et al., 2016; Sinha et al., 2018) and CNNs, (Kim, 2014; Shimura et al., 2018) and character-level CNN Zhangetal. (2015) were applied in text classification problem which improve the accuracy

Introduction of graph network in the text processing explore more rich semantics with the introduction of GCN (Kipf and Welling, 2017) and also Text GCN (Yao et al., 2019). Later attention layer was added to introduce GAT and HGAT network which perform better in less amount of data.

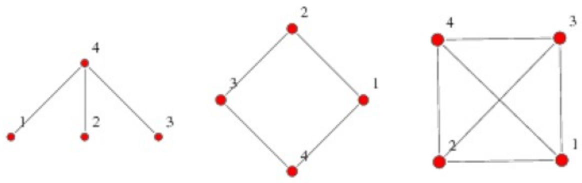
Why Graph Network?

- Natural Representation of Data in graph
  (preserve higher level semantic levels)
- Unlike CNN which has fixed size grid structure (grid graph), it is more flexible
- No fixed ordering
- Node can have any amount of neighbor

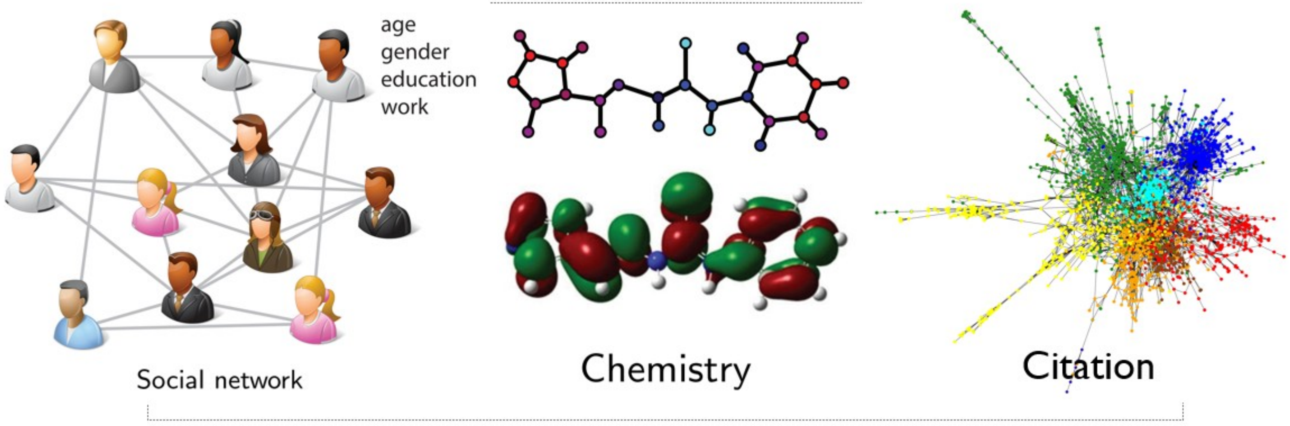


Graph Network

- Graph is collection of nodes and edges between them
- It can represent a real world objects since every object has property and interactions with some other objects.
- Graph (V,E)
- Adjacency matrix



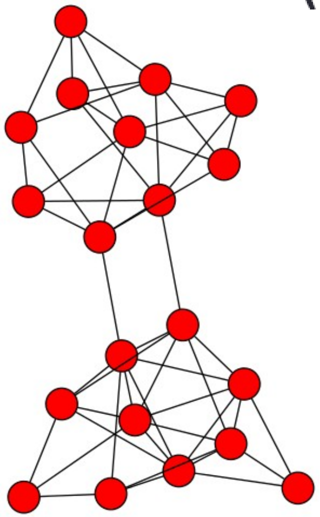
$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$



Data representation in Graph

Network Embedding

G = (V, E)

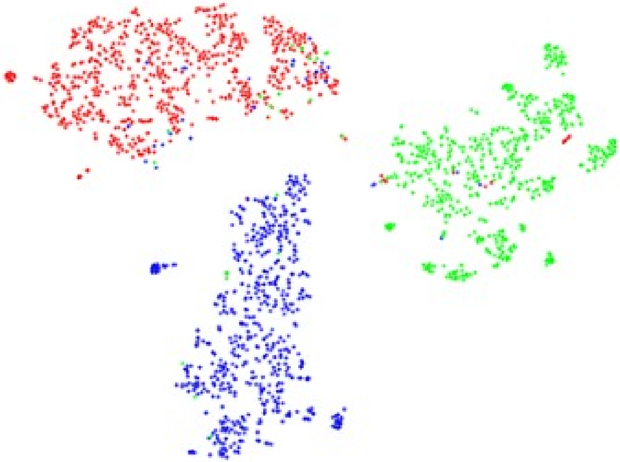


generate

embed

G = (V)

Vector Space



Methodology

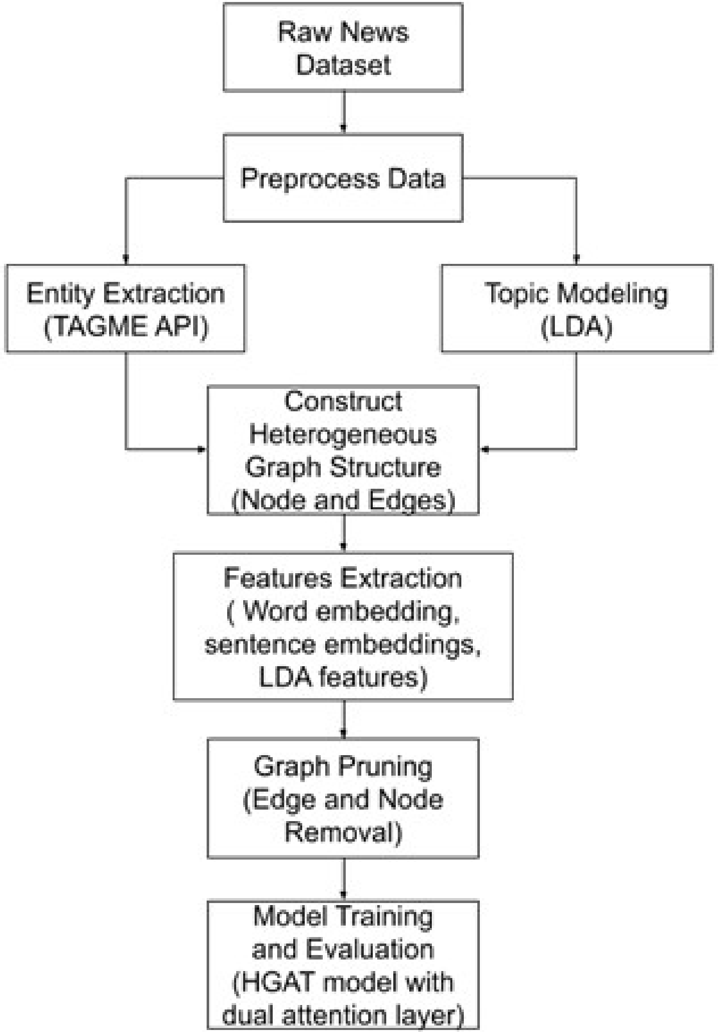
Datasets

AGNews datasets- 6000 data
Derived from base AGNews dataset which contain 4,96,835 categorized news articles from more than 2000 news sources more than 1 year. Updated(09/09/2015)

Format: [index] \t [category] \t [news text] \n
Category value is: 1-World, 2-Sports, 3-Business, 4-Sci/Tech

Index	Category	Document
0	2	lenny wilkens gained some job security despite coaching without hand picked assistant for the first time in his career as the new york knicks defeated the philadelphia ers tuesday night
1	1	police in georgia breakaway abkhazia region refuse to take orders from the government
2	3	ap carrefour the world no retailer said wednesday that its profit increased percent in the first half of the year as robust international operations and lower finance costs helped it ride out tough conditions at its core french superstores
3	2	dale rogers completed percent of his passes committed no turnovers and was the team leading rusher yet the spartans still lost by to stanford

Overall Block Diagram



1. Entity Extraction

- Use of TagME API
- Input one sentence
- Output list of entities in that sentence [{entity 1}, {entity 2},.....]
- Format of output:
  entity 1 = {spot, start, link_probability, rho, end, id, title }

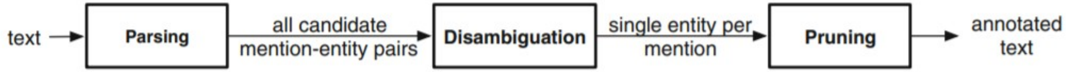


Fig. 1. Annotation pipeline in the TAGME system.

2. Topic Modelling

- Represents abstract topic that occur in collection of documents
- Latent Dirichlet Allocation (LDA) is used for topic modeling

$T_i = (\theta_1, \theta_2 ....., \theta_w)$

where, i is the number of topics

θ_w represents keywords and w is the vocabulary size

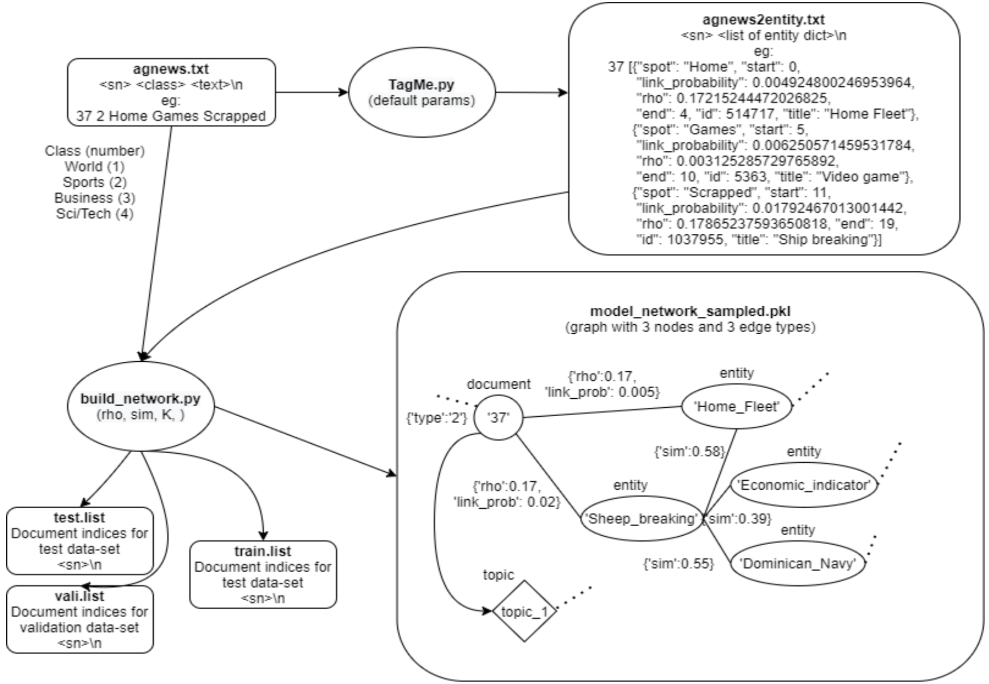
Here each T_i will form a node and each document is assigned with P topics that have the highest probabilities

3. Building Graph

- Graph type: Undirected and Heterogeneous Graph
- Edge has a label to store link strength

- Node type: Document - Ⓓ, Topic - Ⓣ, Entity - Ⓔ
- Edge type: Ⓓ↔Ⓣ
  Ⓓ↔Ⓔ
  Ⓔ↔Ⓔ
- Two documents are linked in following formations with each other
  Ⓓ↔Ⓣ↔Ⓓ
  Ⓓ↔Ⓔ↔Ⓓ
  Ⓓ↔Ⓔ↔Ⓔ↔Ⓓ
- Hyperparameters:
  enitity_similarity : 0.5, min_entity_connection: 10
  num_train_text_class: 40-600

Overall Process of Building Graph



- Document node and it's property ('type') is extracted from agnews.txt
- Document to Entity edge and it's properties ('rho', 'link_prob') are extracted from agnews2entity.txt
- Entity to Entity edge along with its property ('sim') is found by using gensim word2vec similarity measure.
- Topic Document edge is created based on the highest probability from LDA model top 3 topics in a document are connected

4. Feature Representation

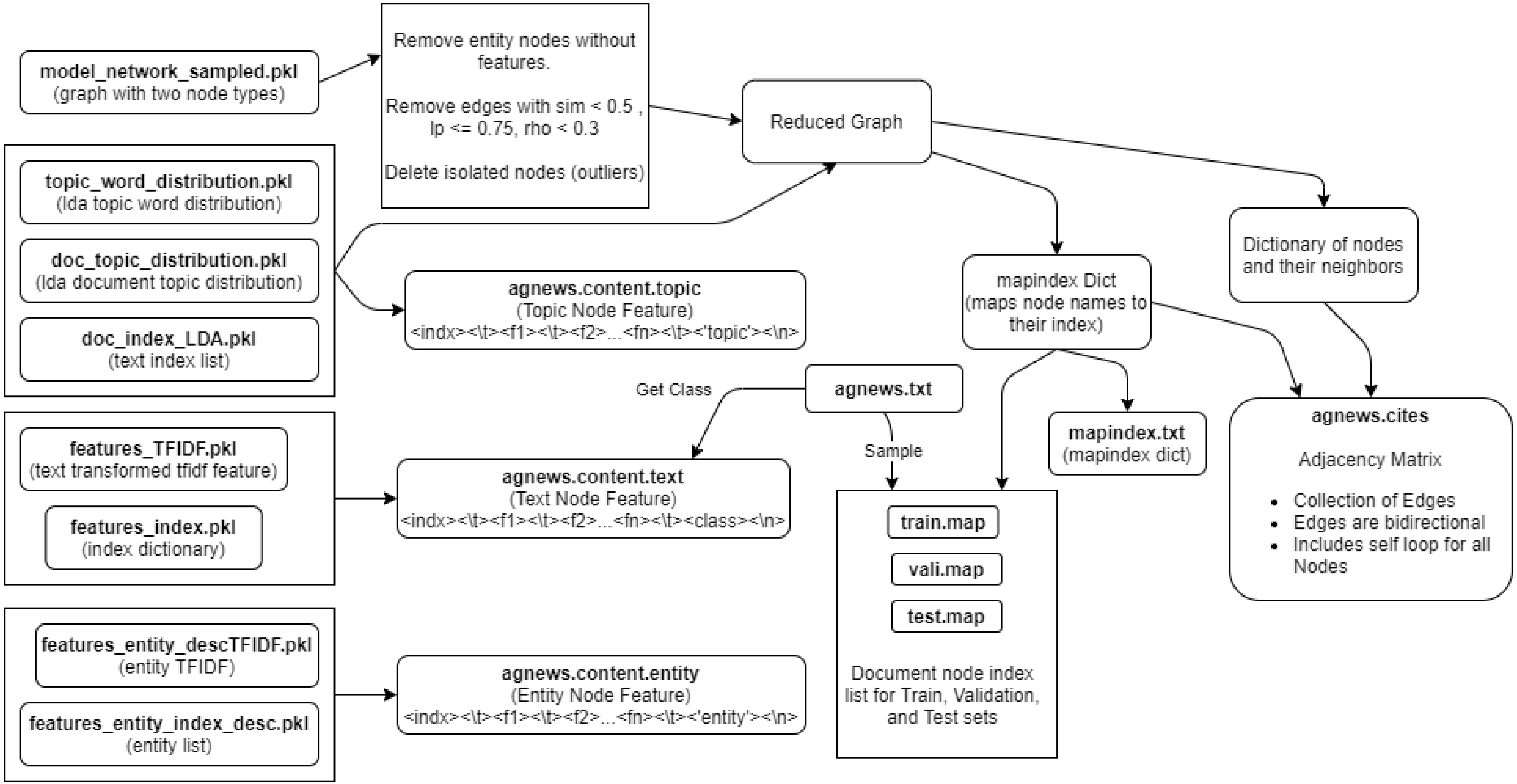
- Building the entity node features
  Pre-trained word2vec model is used to get the word embedding of the entity

- Build Text node features
  Sentence embeddings is generated by averaging the word embedding of each word in the text

- Build Topic node features (LDA components), Obtained from topic model

- Basic preprocessing used
  - Tokenize
  - Lowercase
  - remove non-alpha characters
  - remove stop words
  - word that repeats less than 5 (whole data)

5. Graph Pruning (Overall Diagram)



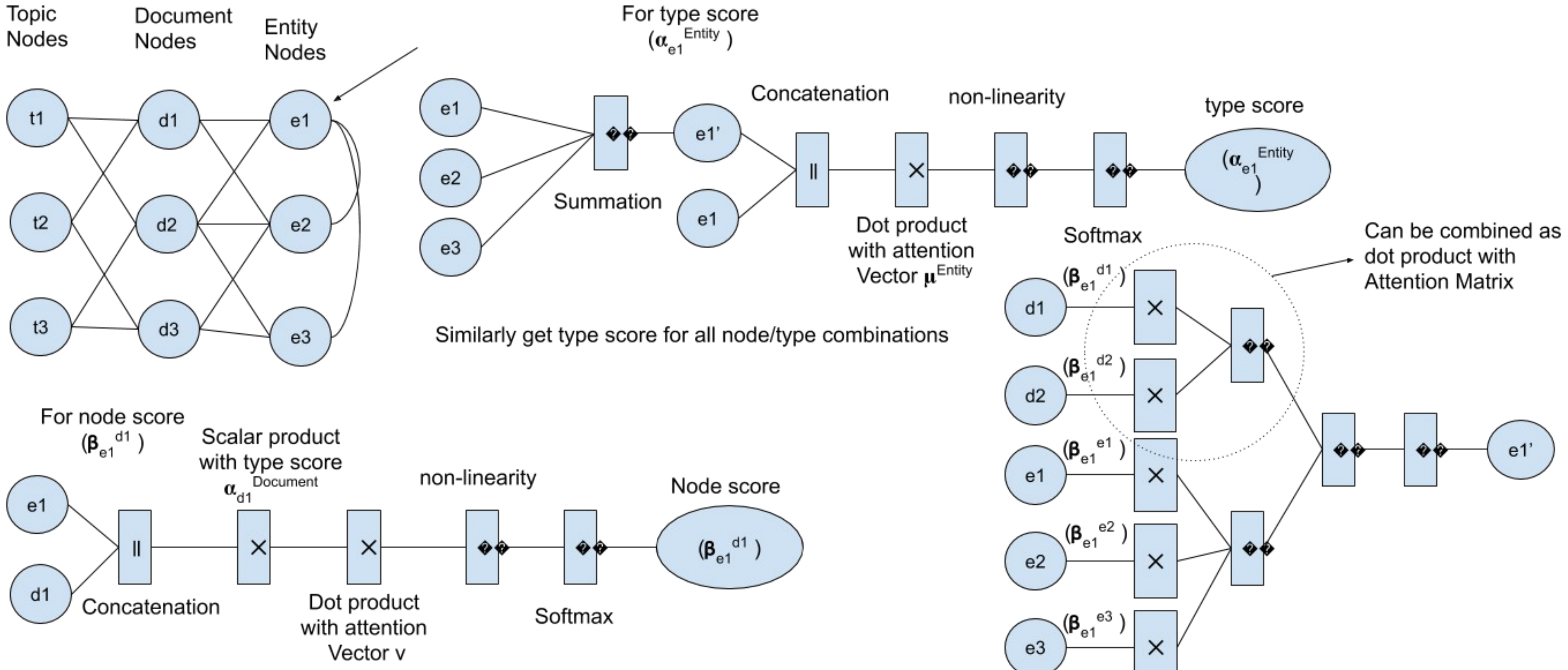
4. Training

HGAT model
HIdden Unit = 512
Class = 4
Dropout Rate (DP) = 0.95
Gamma = 0.1

Adam Optimizer
Learning Rate (LR) = 0.005
Weight Decay (WD) = 5e - 8

Epoch = 300
Seed = 42

HGAT model



Similarly get node score for all node pair combinations

Attention in HGAT

- Dual Attention Layer

- Type level attention and Node Level attention

- Type Level concentrate on which node to give more attention based on its type

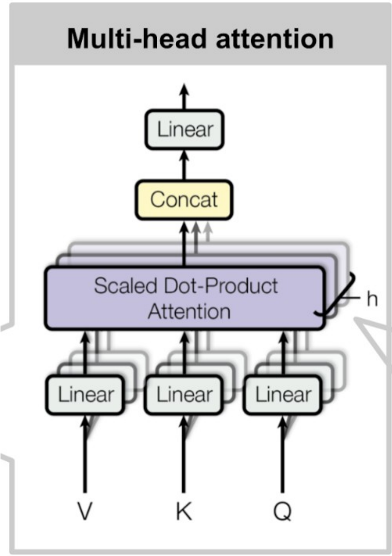
- Node Level attention is used to focus on attention in individual node

$$\alpha_{\mathrm{T}} = \sigma(\mu_{\tau}^{\mathrm{T}} . [h_{v'} || h_{\mathrm{T}}])$$

$$\alpha_{\mathrm{T}} = \frac{exp(\alpha_{\mathrm{T}})}{\sum_{\tau' \epsilon \mathrm{T}} exp(\alpha_{\tau'})}$$

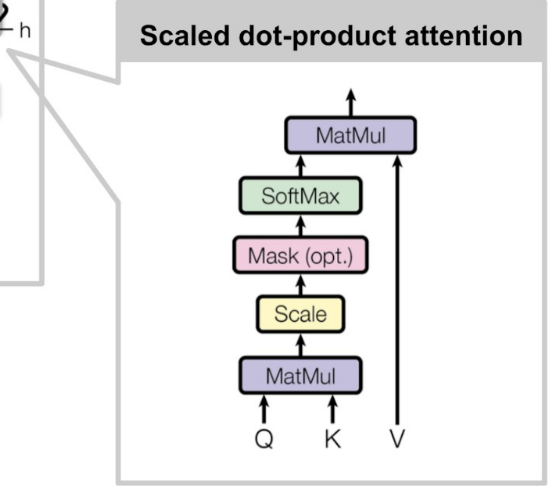
α_{T} is the type level attention which is then passed through softmax layer

Multi-head Attention Network

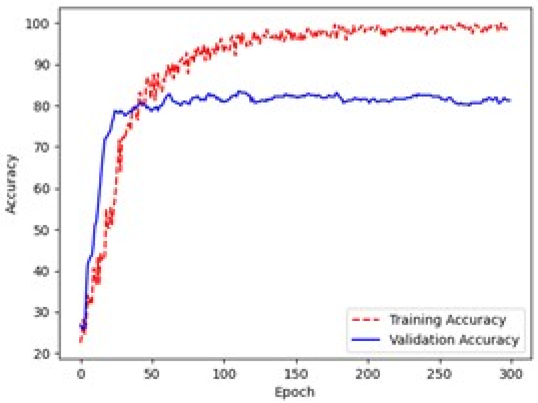


Zoom-In!

$$Attention(Q, K, V) = softmax(\frac{Q\,\boldsymbol{K}^T}{\sqrt{n}})\mathrm{V}$$



Results



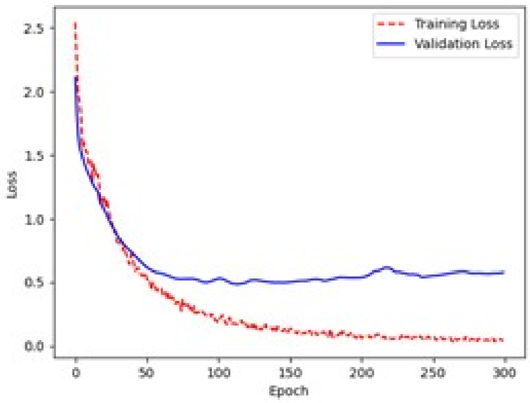


Fig: Accuracy Curve and Loss Curve

Comparison table

Datasets	Document Node	Entity Node	Topic Node	Training Nodes	Validation Nodes	Test Node
AGNews	6000	10298	15	160	160	5680

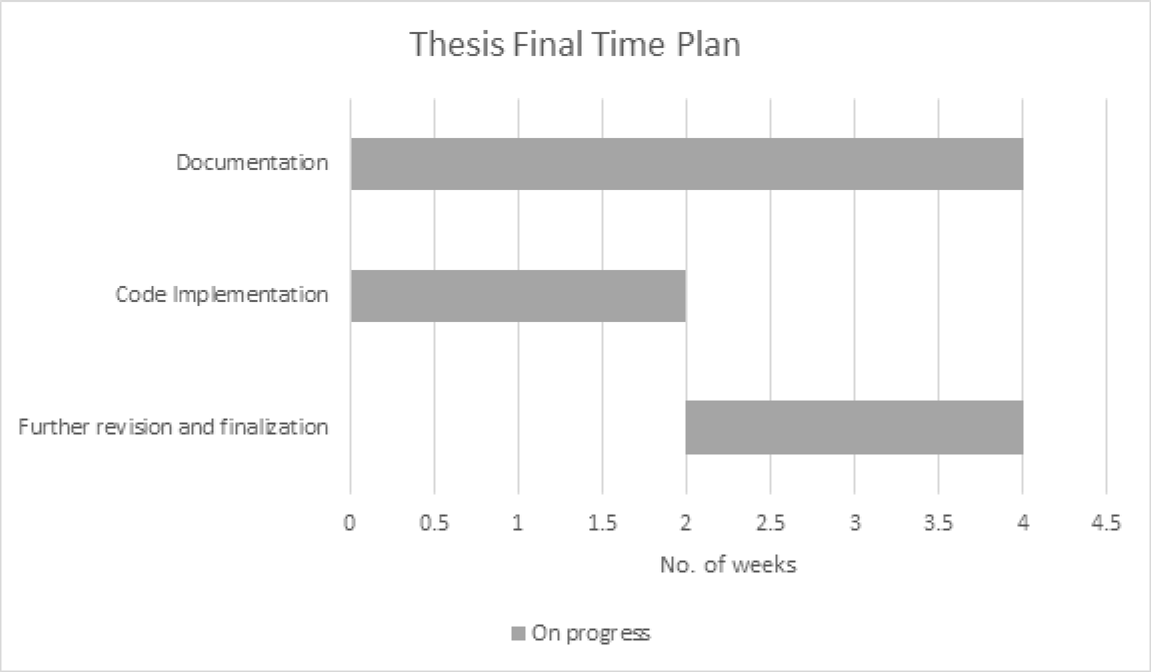
Model \ Metrices	Accuracy	Precision	Recall	F1 Score
HGAT base paper	72.10	-	-	-
HGAT*	**76.30**	**76.4**	**76.3**	**76.3**

HGAT* applied in this research

Work Remaining

- Optimization of model by parameter tuning
- Custom Word2vec training with available data to observer the performance
- Testing with other text datasets (another news dataset with more classes)
- Application of Doc2Vec model in text embeddings
- Improvement and update of documentation
- Variational Autoencoders for topic modeling

Gantt Chart



Thesis Final Time Plan
Documentation
Code Implementation
Further revision and finalization
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
No. of weeks
On progress

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Thank You